Predictive Analysis of Hospital Overall Ratings Using Healthcare Quality and Safety Indicators

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***Abstract*— Star ratings of hospitals are a composite summary of several dozens of measures of quality that stakeholders depend on, however, it is hard to understand how particular measures contribute to it. This paper explores to what extent seven national comparison groups of quality and safety indicators as well as four attributes of hospitals can predict star ratings of hospitals in the United States. After cleaning the publicly available Hospital Information dataset (4,812 records), 3,567 hospitals having a proper 1-5 star rating were kept. A composite Healthcare Quality-Safety Score was created from the indicator groups, and three class-balanced classifiers – multinomial Logistic Regression, Decision Tree and Random Forest were trained within a leakage-free preprocessing pipeline and tested on a stratified 20% hold-out sample. Logistic Regression and Random Forest showed 66.4% and 66.1% accuracy (weighted F1 of 0.678 and 0.673 respectively) against 58.8% of the Decision Tree and 49.4% of a majority class classifier. The composite score was the most important feature and improved Random Forest accuracy from 64.3% to 66.1%. Predictions were mostly made for adjacent star ratings and extremely low/high ratings were still challenging.**

Index Terms—hospital rating prediction, healthcare quality indicators, feature engineering, random forest, class imbalance, multiclass classification.

I. Introduction

The choice of a hospital is no small feat, and one of the most recognizable aids to the process is the Overall Hospital Quality rating of 5 stars issued by the U.S. Centers for Medicare and Medicaid Services (CMS). This rating summarizes 57 measures across seven domains: mortality, safety of care, readmission, patient experience, effectiveness of care, timeliness of care and efficient use of medical imaging, into one to five stars [1]. At the same time, hospitals use the rating to guide their quality improvement efforts, so it is important for both consumers and providers to understand how it works.

In spite of the prominence, the rating has drawn criticisms before. Analysis of the properties of the rating found summary scores to be concentrated around the center of the scale, and the star rating not being a reliable proxy for performance on condition-specific measures [2]. However, open source tools make it possible for researchers to recomputed and predict star ratings based on publicly available Care Compare data [3]. While it demonstrates the ability to reproduce the rating from its ingredients, it gives little information about the amount of signal conveyed through coarse groups of indicators available from consumer-oriented datasets.

This

II. Related Work

Analysis of hospital rating. Kurian et al. built an ordered logistic regression model on 4,519 hospitals, applied stepwise selection and multiple imputation, and demonstrated that twenty CMS measures contained all the necessary information for predicting the star rating [1]. Hu and Nerenz demonstrated that overall scores have very narrow distribution range and are not significantly related to specific disease-related quality measures in California and New York [2]. Locey et al. provided an open source tool for estimation of ratings with 99.6% accuracy six months prior to their public release using the CMS methodology and corresponding data [3]. The above-mentioned studies operate with precise numeric measures; we are going to examine the information capacity of coarse national-comparison categories.

Application of machine learning to healthcare. The reviews on artificial intelligence in medicine emphasize increase of non-image data usage and discuss issues such as lack of data and biases [4]. They also provide an overview of multimodal [5] and self-supervised [6] learning in biomedical field. Our problem is rather traditional structured data prediction task and thus scalability is less critical than model selection and interpretability.

Benchmark results in tabular data. Shwartz-Ziv and Armon demonstrated that XGBoost algorithm outperforms several recent deep models and

Interpretability and Evaluation. Ghassemi et al. warn against the use of post hoc explanation techniques that cannot be trusted to give useful patient-level decision support and recommend thorough validation instead [12], while Markus et al. review explainability options for designing trustful healthcare AI systems [13] and Rudin et al. make the case for interpretability in high-risk contexts [14]. We adhere to this advice by comparing a linear model to a shallow tree and a forest, as well as by stating importance only as an association. Chicco and Jurman have shown that ROC AUC is misleading and that metrics that incorporate the whole confusion matrix, like MCC, are to be trusted in binary classification problems [15]; we report per-class precision and recall and the confusion matrix alongside the usual weighted metrics.

III. Dataset and Preprocessing

The Hospital Information dataset (HospInfo.csv, Kaggle) consists of 4,812 rows and 29 columns, including IDs and address columns, hospital type, ownership, emergency services and EHR use indicator, the overall rating of the hospital, and seven national comparison measures (Mortality, Safety of Care, Readmission, Patient Experience, Effectiveness of Care, Timeliness of Care, Efficient Use of Medical Imaging), along with their respective footnotes. There are no duplicate rows in the dataset. Footnote columns have 55%-79% empty rows and thus are not considered. The EHR flag is missing in 144 observations, imputed in the pipeline and has just one distinct value (true), therefore, providing no discrimination power.

The target variable is Hospital overall rating. The records, which do not have the rating value from 1 to 5 (rating not

available), were filtered out, resulting in 3,567 rows. The target distribution (Table I, Fig. 1) is skewed towards three stars, with only 107 one-stars and 82 five-stars hospitals, motivating class-weighting and per-class evaluation

.TABLE I
CLASS DISTRIBUTION AFTER CLEANING (N = 3,567)

Stars	**1**	**2**	**3**	**4**	**5**
Hospitals	107	678	1,761	939	82
Share (%)	3.0	19.0	49.4	26.3	2.3
Train	86	542	1,408	751	66
Test	21	136	353	188	16

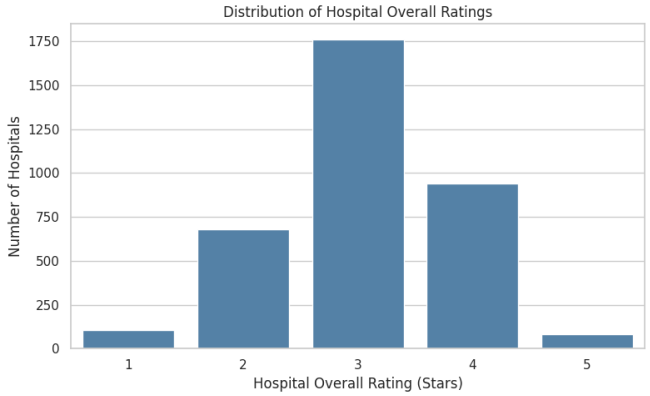


Fig. 1. Distribution of hospital overall ratings (notebook output).

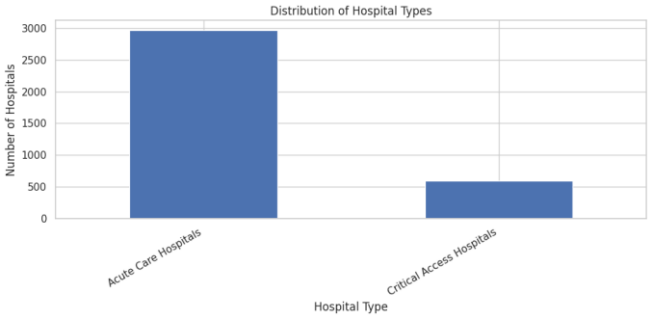


Fig. 2. Distribution of hospital types in the cleaned data.

The exploratory analysis of the clean dataset (Figs. 2–4) reveals that there are two hospital types, 3,000 acute care hospitals and around 600 critical access hospitals, and there are ten ownership types, of which the largest is voluntary not-for-profit (private), while the smallest are physician, federal and tribal hospitals; “three stars” is the most common rating among all major ownership types. Among the seven indicator groups, the most common label is “same as the national average”; the patient experience indicators are distributed almost equally into three categories (same, above, below), while “Not Available” label is common for “Safety of Care” and “Efficient Use of Medical Imaging” (around 900 and 800 hospitals correspondingly) but uncommon for “Readmission”.

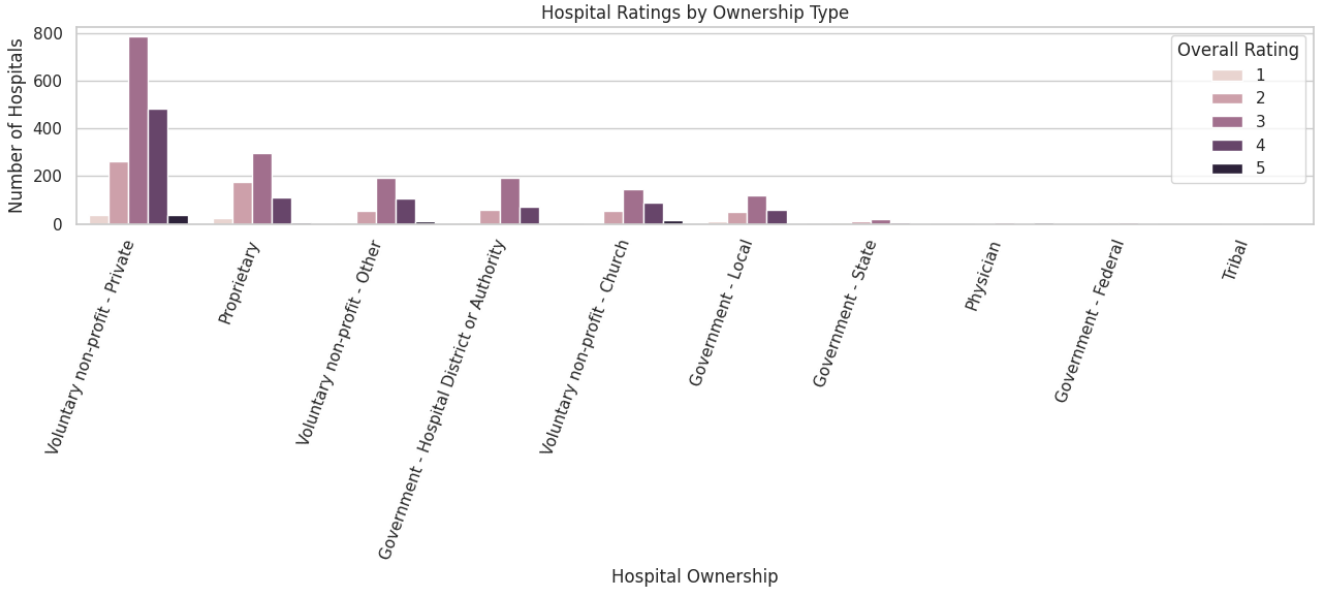


Fig. 3. Hospital ratings by ownership type.

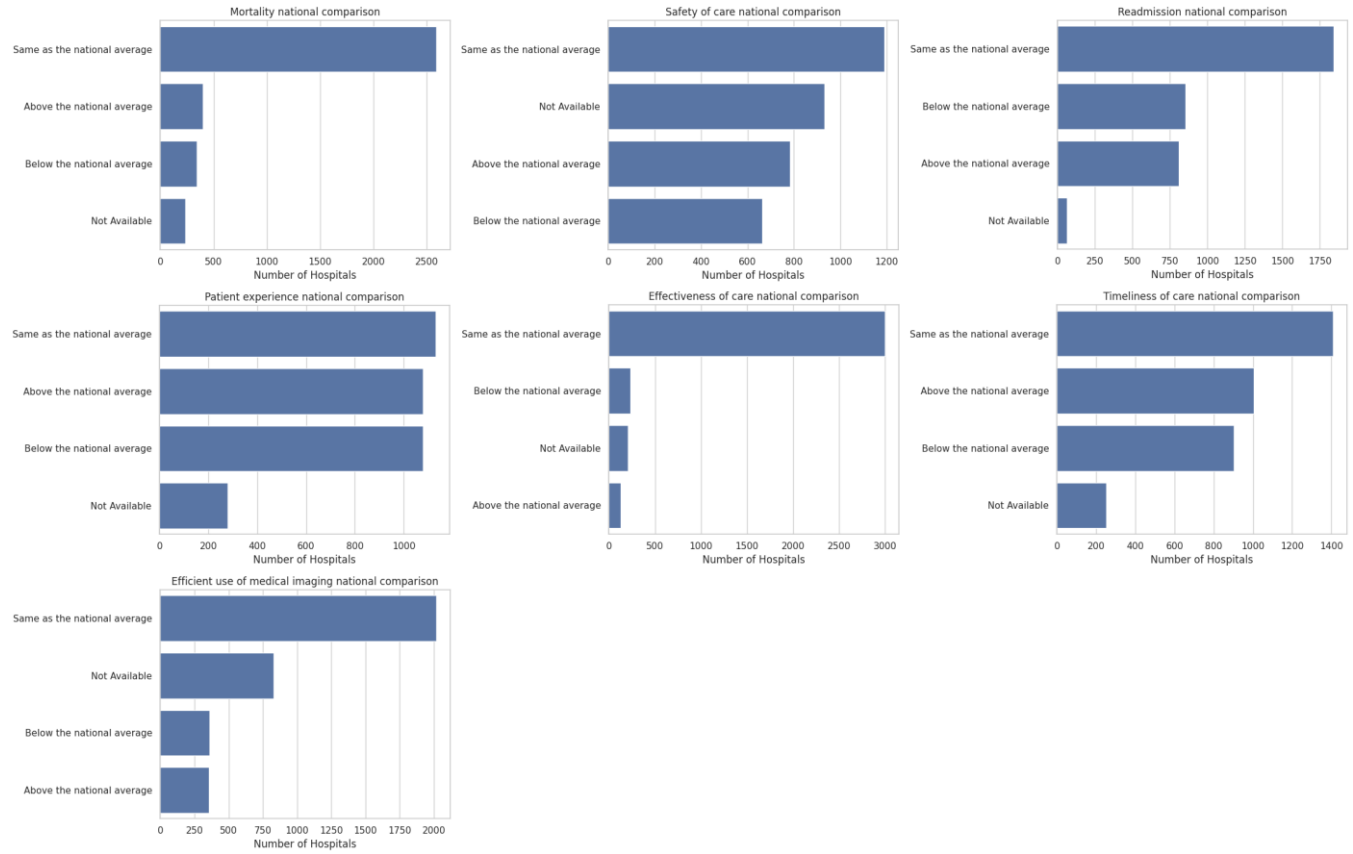


Fig. 4. Distribution of the seven national-comparison indicator groups (notebook subplots, re-tiled).

IV. Methodology

A. Pipeline Overview

Fig. Summary of 5 Workflow

Data after cleaning is split into two parts. Preprocessing model will be fitted using training set only and three classification models will use common pre-processing model to ensure that differences in their performance is because of learning algorithm itself..

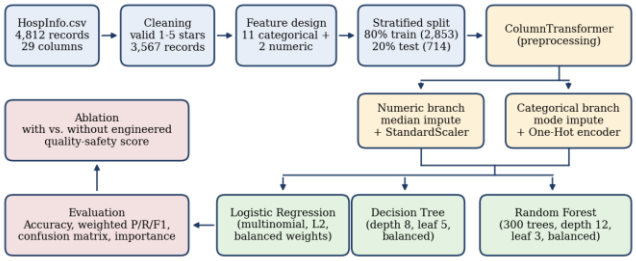


Fig. 5. Proposed end-to-end methodology and model architecture.

B. Feature Engineering

Each national-comparison label was mapped to a score $s_{ik} \in \{+1, 0, -1\}$ for "above", "same as" and "below the national average"; "Not Available" remains missing. For hospital i with available indicator set K_i, the composite score and the indicator count are

$$S_i = (1 / |K_i|) \Sigma_{k \in K_\{i} \} s_{ik} \quad (1)$$

$$n_i = |K_i|, \quad K_i = \{ k : s_{ik} \text{ is observed } \} \quad (2)$$

The score summarises overall relative performance in [−1, 1], while n_i lets the model discount hospitals whose score rests on few indicators (up to seven are available per hospital). The final feature set (Table II) has 13 inputs: four categorical hospital attributes (type, ownership, emergency services, EHR meaningful use), the seven categorical indicator labels, and the two numeric features S_i and n_i.

TABLE II
Input Features Used in the Models

Feature	Type	Values
Hospital Type	Categorical	Acute care; critical access
Hospital Ownership	Categorical	10 ownership categories
Emergency Services	Categorical	True / False
Meets EHR meaningful-use criteria	Categorical	True only (144 missing)
Seven national-comparison indicators	Categorical	Above; same; below; not available
Quality–Safety Score S_i	Numeric	[−1, 1]; missing if no indicator
Available indicator count n_i	Numeric	Integer, up to 7

C. Preprocessing

Numeric features are median-imputed and standardised, and categorical features are mode-imputed and one-hot encoded with unseen categories ignored:

$$z = (x - \mu_{train}) / \sigma_{train} \quad (3)$$

$$o_j(x) = 1[x = c_j], \quad j = 1, \ldots, m \quad (4)$$

Both steps live in a *ColumnTransformer* inside each model pipeline, so test data never influence imputation values, scaling statistics or category sets.

D. Classification Models

Logistic Regression is the multinomial (softmax) model with L2 regularisation and class-balanced weights:

$$P(y = c \mid x) = exp(w_c^T x + b_c) / \Sigma_{r=1}^{5} exp(w_r^T x + b_r) \quad (5)$$

$$L = -\Sigma_i \omega_{y_\{i}\} \log P(y_i \mid x_i) + (1 / 2C) \| W \|_2^2 \quad (6)$$

where the balanced weight of class c is

$$\omega_c = N / (K \cdot n_c) \quad (7)$$

with N training samples, $K = 5$ classes and n_c samples in class c. The *Decision Tree* grows by choosing, at each node, the split that most reduces the weighted Gini impurity

$$G(t) = 1 - \Sigma_c\, p_c(t)^2 \quad (8)$$

subject to a maximum depth and a minimum leaf size. The *Random Forest* [16] averages $T = 300$ such trees grown on bootstrap samples with random feature subsets, and predicts by majority vote,

$$\hat{y}(x) = \mathrm{argmax}_c\, \Sigma_{t=1}^{T}\, 1[h_t(x) = c] \quad (9)$$

and its impurity-based importance of feature f is the average Gini decrease it produces over all splits and trees.

E. Evaluation Metrics

For each class, precision $P_c = \mathrm{TP}_c/(\mathrm{TP}_c+\mathrm{FP}_c)$, recall $R_c = \mathrm{TP}_c/(\mathrm{TP}_c+\mathrm{FN}_c)$ and $F1_c = 2P_cR_c/(P_c+R_c)$ are computed. Aggregate scores are support-weighted:

$$Acc = (1/N_{test})\, \Sigma_i\, 1[\hat{y}_i = y_i] \quad (10)$$

$$M_w = \Sigma_c\, (n_c^{test} / N_{test}) \cdot M_c, \quad M \in \{P, R, F1\} \quad (11)$$

F. Testing Environment and Hyperparameters

All experiments ran in Google Colab (hosted Python 3 runtime) using pandas, NumPy, scikit-learn [17], Matplotlib and Seaborn; library versions are those preinstalled in Colab at run time. The data were split 80/20 with stratification on the rating and a fixed random seed of 42, giving 2,853 training and 714 test records. Hyperparameters were fixed *a priori* (Table III); no search or cross-validation was run, and unlisted parameters use scikit-learn defaults (for example $C = 1.0$ with the L2 penalty for Logistic Regression).

TABLE III
MODELS AND HYPERPARAMETERS

Model	Settings
Logistic Regression	max_iter = 2000; class_weight = balanced; random_state = 42
Decision Tree	max_depth = 8; min_samples_leaf = 5; class_weight = balanced; random_state = 42
Random Forest	n_estimators = 300; max_depth = 12; min_samples_leaf = 3; class_weight = balanced; n_jobs = −1; random_state = 42
Preprocessing	Numeric: median impute + StandardScaler; categorical: most-frequent impute + OneHotEncoder (handle_unknown = ignore)

V. RESULTS AND DISCUSSION

A. Model Comparison

Table IV and Figure 6 show test set performance for the 714 hold out hospitals. The Logistic Regression model is slightly superior to Random Forest in terms of all aggregation metrics (accuracy is only 0.003 lower; this difference is equivalent to two hospitals), while the Decision Tree model lags behind both of the other models. All three models significantly outperform the majority class baseline, which always predicts three stars and achieves 353/714 = 49.4% accuracy. Since the two top-performing models perform roughly equally, the Random Forest model was chosen for the error and importance analyses due to the availability of built-in feature importances

.TABLE IV
TEST-SET PERFORMANCE (WEIGHTED AVERAGES)

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.6639	0.7141	0.6639	0.6777
Random Forest	0.6611	0.6986	0.6611	0.6732
Decision Tree	0.5882	0.6604	0.5882	0.6089
Majority class (derived)	0.4944	—	—	—

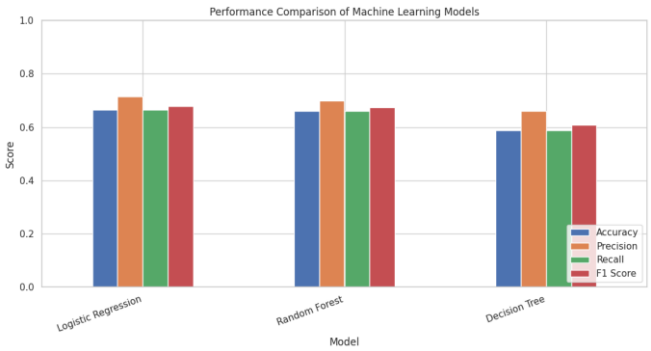


Fig. 6. Performance comparison of the three models.

B. Error Analysis

The Random Forest confusion matrix (Fig. 7) shows that 472 of 714 hospitals (66.1%) lie on the diagonal and that errors are almost always one star away: for example, 67 of 353 three-star hospitals were predicted as four stars and 40 as two stars, while only 2 were predicted as one star and 4 as five stars; just 8 of the 242 errors span two stars. This pattern is expected because the indicator categories are ordinal summaries of the same quality dimensions that produce the rating. The per-class report (Table V) supports this reading: F1 is 0.74 for three stars, 0.67 for four and 0.59 for two stars, but only 0.34 for one star (precision 0.25, recall 0.52) and 0.42 for five stars (precision 0.29, recall 0.75). Balanced class weights therefore buy high recall on the rare extremes at the price of low precision, and the macro-averaged F1 is 0.55 against a weighted 0.67. Using weighted scores alone would hide this weakness, in the spirit of [15].

TABLE V
RANDOM FOREST PER-CLASS TEST RESULTS

Class (stars)	Precision	Recall	F1	Support
1	0.25	0.52	0.34	21
2	0.60	0.58	0.59	136
3	0.81	0.68	0.74	353
4	0.65	0.69	0.67	188
5	0.29	0.75	0.42	16
Macro average	0.52	0.65	0.55	714
Weighted average	0.70	0.66	0.67	714

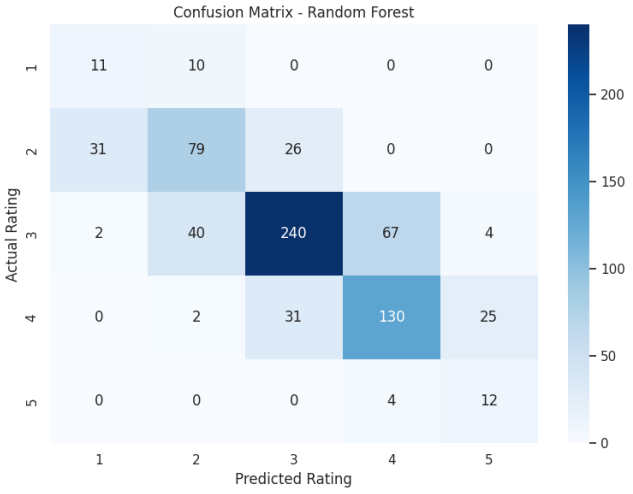


Fig. 7. Confusion matrix of the Random Forest on the test set.

C. Feature Importance

The engineered Healthcare Quality–Safety Score has by far the largest impurity-based importance (0.238; Table VI), followed by the patient-experience categories “below” (0.081) and “above” (0.076), the readmission categories (0.062, 0.054,

0.053) and the safety-of-care "below" category (0.052) (Fig. 8). Timeliness and mortality categories contribute less than patient experience and readmission, and hospital type, ownership and emergency services do not appear among the top fifteen. Importance values describe association within this model, not causal effect, and impurity-based importance can favour features with many split opportunities; they should be read as a guide to where the signal is rather than a policy recommendation, consistent with the cautions in [12] and [14].

TABLE VI
TOP TEN RANDOM FOREST FEATURE IMPORTANCES

Rank	Feature	Importance
1	Quality–Safety Score (engineered)	0.2380
2	Patient experience: below average	0.0815
3	Patient experience: above average	0.0765
4	Readmission: below average	0.0619
5	Readmission: above average	0.0539
6	Readmission: same as average	0.0525
7	Safety of care: below average	0.0523
8	Timeliness of care: below average	0.0363
9	Safety of care: above average	0.0331
10	Mortality: above average	0.0234

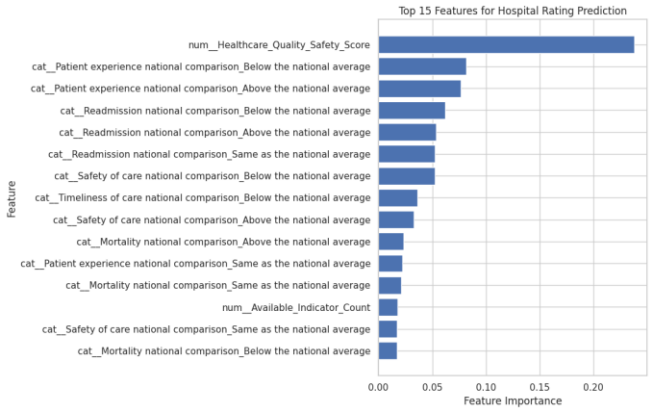


Fig. 8. Top 15 Random Forest features for rating prediction.

D. Ablation of the Engineered Score

To measure what the engineered features add, the same Random Forest was trained on the original 11 categorical inputs and on the full 13-feature set (Table VII). The engineered score and count raise accuracy by 1.8 points and weighted F1 by 1.8 points. This is a single-split comparison with identical seed and hyperparameters, so it indicates a consistent benefit but not a statistically tested one.

TABLE VII
EFFECT OF FEATURE ENGINEERING (RANDOM FOREST)

Feature set	Accuracy	F1
Original 11 categorical features	0.6429	0.6555
+ Quality–Safety Score and count (13)	0.6611	0.6732

E. Comparison with Benchmarks from the Literature

Table VIII compares the current results to the methods used in the references. Comparison of figures is not feasible due to the fact that the studies employ different data versions, performance metrics, or objectives. The focus of this comparison is therefore on design considerations and implications for the current study..

TABLE VIII
COMPARISON WITH REFERENCE BENCHMARKS

Study	Method / data	Relevance to this work
Kurian [1]	Ordered logistic regression; 4,519 eligible hospitals; numeric CMS measures	Linear model baseline; our multinomial LR is analogous but uses coarse categories
Locey [3]	Reproduction of CMS method; 99.6% accuracy six months ahead	Upper bound when full measure data are available; ours uses 7 categories only
Shwartz-Ziv [7]; Grinsztajn [8]; Borisov [9]	Tree ensembles vs. deep models on tabular data	Supports tree and linear models on ~3.6k samples
Zabërgja [11]	68-dataset study favouring newer deep models	Counterpoint; deep models left for future work
This work	LR / DT / RF; 13 features; 3,567 hospitals	LR 0.664 acc., RF 0.661, DT 0.588, majority 0.494

Two clear lessons arise from this. First, tree-based methods and linear models are equally effective for smaller sized tabular datasets [7] – [9], and here, a linear model with regularization is just as good as a forest with 300 trees, indicating that the ordinal relationship of the attributes is fairly linear. Second, the difference between our 66% and near perfect score from [3] is likely due to the availability of information; that is, the CMS score is calculated using precise numeric values, while here we can only observe the domains being above/below the average level.VI. LIMITATIONS AND FUTURE WORK

The results are from one stratified partitioning without cross-validation or hyperparameter tuning, thus differences of a few tenths of a point like that of Logistic Regression and Random Forest do not necessarily mean much. The 1,245 hospitals that had no rating were omitted from the dataset, which may have skewed the sample, and this is only one version of the data available from one U.S. release. The engineered feature since it is constructed from the same labels of indicators contains no information besides the compression. Possible further improvements: stratified k-fold cross-validation, ordinal and cost-sensitive models taking into account order of stars, gradient-boosted decision trees, probability calibration, SHAP explanations [12] of the local kind and using numeric indicators for testing the information-loss hypothesis..

VII. CONCLUSION

The pipeline was designed and analyzed in order to create a pipeline that is completely devoid of any form of leakage to predict the U.S. Hospital Overall Star Ratings from coarser categories of quality and safety indicators. The results obtained from Logistic Regression (Accuracy 66.4%, F1 score 0.678) and Random Forest (Accuracy 66.1%, F1 score 0.673) indicate a clear superiority over the decision tree model and a majority class baseline. Errors occur primarily between consecutive star ratings, and the healthcare quality and safety score becomes the most significant feature as well as an improvement in accuracy

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